



END EVAL

GANFORCE

PROJECT ₁



Introduction to the Project

- What is Style Transfer?

- Applying the "style" (colors, textures, brushstrokes) of one image to the "content" of another.

- Why is it exciting?

- Merges art with AI, generates creative visuals, can imitate famous painters.

- Real-world Applications:

- Art generation, photo filters, gaming, film post-production.

Project Objective

- To perform artistic style transfer using CycleGAN.
- Transfer styles of famous painters (e.g., Monet, Van Gogh) to everyday photos.
- Train a model on unpaired datasets to perform image-to-image translation.

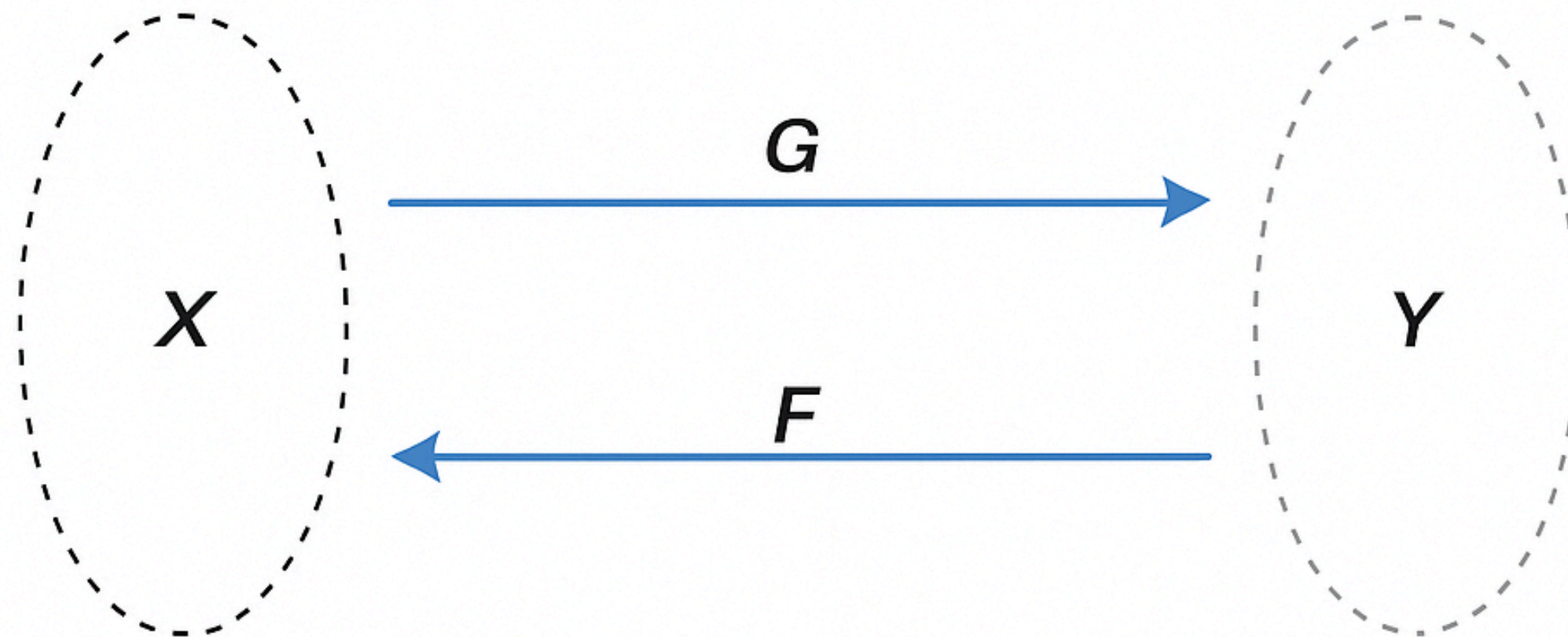
Cycle GAN

CycleGAN (Cycle-Consistent Generative Adversarial Network) is a type of deep learning model used for image-to-image translation — converting images from one domain to another without requiring paired training data.

In general , if we have a set of regular photos and a set of Picasso-style images , then after training CycleGAN , we can input a photo and get its Picasso-ized version as ouput



Framework of CycleGAN



Two Generators:

$G: X \rightarrow Y$ (It converts images from domain X to domain Y)

$F: Y \rightarrow X$ (It converts images from domain Y to domain X)

Two Discriminators:

D_Y : distinguishes real Y images from fake ones generated by G

D_X : distinguishes real X images from fake ones generated by F

Discriminator D_X

Discriminator D_X



Cycle GAN

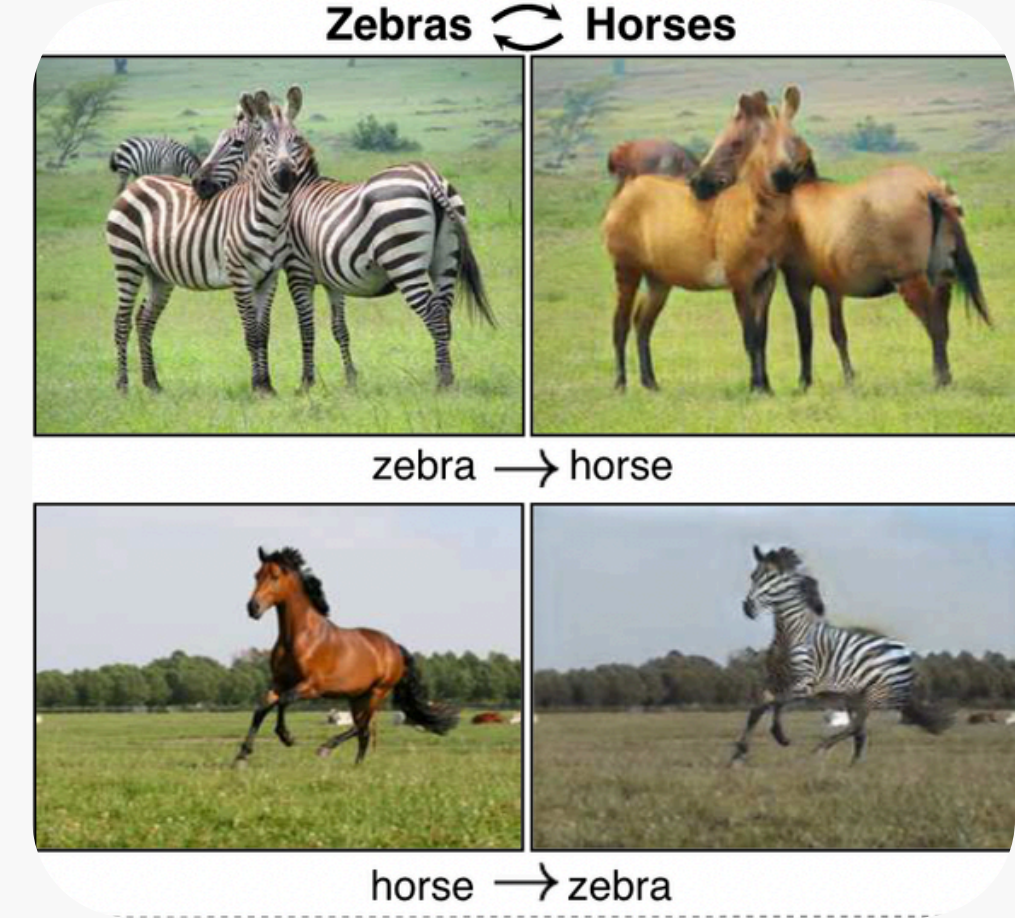
Example using Zebra and Horse:

If we input a brown horse in a field

Then Generator G changes the horse's skin pattern and colour to look like a zebra like stripes appear but background and body structure remains same

The Generator F then takes this fake zebra and converts it back into horse

If we observe here CycleGAN never saw horse-zebra pairs. It just had some random horse and zebra photos and yet it learned the style difference without the changing the content (pose , shape)



Application of cycle gans

- *Image style transfer*
- *Domain adaptation*
- *Medical imaging*
- *Satellite and Map image conversion*
- *Object transfiguration*
- *Super resolution and restoration*



Source

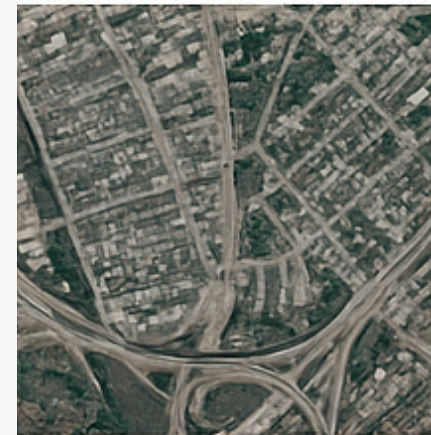


Image style transfer

Target



Domain adaptation



Satellite and map
image conversion



Object transfiguration



Super resolution and
restoration



Super resolution
and restoration

loss function of cycle GAN

The loss function of CycleGAN combines multiple parts that work together to ensure the model does two things well:

- 1. Fool the discriminators (like regular GANs)*
- 2. Preserve content and structure during translation (cycle consistency)*

The CycleGAN Loss Function = Adversarial Loss + Cycle Consistency Loss

The Adversarial loss part ensures that the generated image looks real and the Cycle consistency loss ensures that even though we don't have paired data, the model learns reversible transformations.

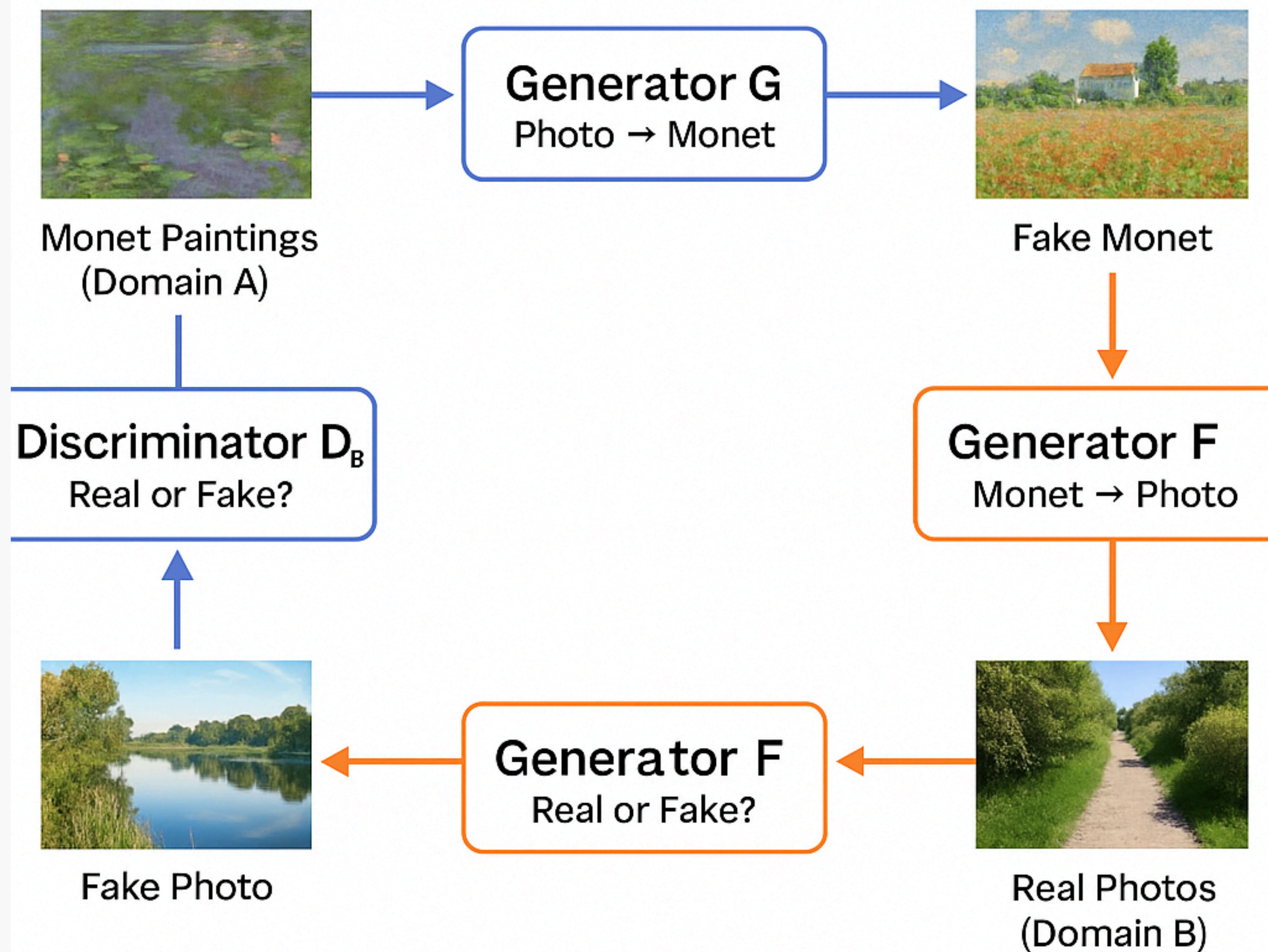
In our project, CycleGAN:

- *Learns the visual mapping between two domains: Monet paintings (Domain A) and real photos (Domain B).*



- *Maintains structure and identity of the original photos using cycle-consistency and identity loss.*
- *Generates high-quality Monet-style artwork from ordinary photographs.*

Use of CycleGAN in Our Project



Code implementation

1. Cloned the official CycleGAN GitHub repository.
2. Installed dependencies via `requirements.txt` and manual installation of `dominate` and `visdom`.
3. Downloaded the `monet2photo` dataset.
4. Trained the model using:

```
!python train.py --dataroot ./datasets/monet2photo --name  
monet2photo_cyclegan --model cycle_gan
```

5. Alternatively, used a pretrained model for testing:

```
!bash ./scripts/download_cyclegan_model.sh monet2photo  
!python test.py --dataroot ./datasets/monet2photo --name  
monet2photo_pretrained --model test --no_dropout
```

Results

The generated Monet-style images from real-world photos showed high visual quality



Conclusion

This project demonstrated how CycleGANs can achieve high-quality artistic style transfer without the need for paired datasets. By leveraging adversarial training and cycle consistency, we successfully transformed ordinary photographs into artwork in the style of Claude Monet. This has broad implications for creative AI and media generation.



Thank you

